

Tasks on Identity and Access Management – IAM

1. Create an IAM user and assign the EC2 and S3 full access role.

Purpose:

STEPS TO CREATE AN IAM USER AND ASSIGN EC2 AND S3 FULL ACCESS:

Navigate to IAM Dashboard

- Login to AWS Console → Search for "IAM" → Click "Users"

Create User

- Click the "Create user" button
- Enter username: <user_name>
- Select "Provide user access to AWS Management Console"
- Provide console access → I want to create an IAM user
- Console password selection: Autogenerated or Custom
- Click "Next"
-

The screenshot shows the 'Specify user details' step in the AWS IAM console. On the left, a progress bar indicates four steps: Step 1 (Specify user details, active), Step 2 (Set permissions), Step 3 (Review and create), and Step 4 (Retrieve password). The main content area is titled 'Specify user details' and contains the following sections:

- User details:** A text input field for 'User name' contains the value 'Omer'. Below the field, a note states: 'The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)'.
- Provide user access to the AWS Management Console - optional:** A checkbox is checked. A note below it says: 'If you're providing console access to a person, it's a best practice to manage their access in IAM Identity Center.'
- Are you providing console access to a person?:** This section contains a 'User type' subsection with two radio button options:
 - ☐ Specify a user in Identity Center - Recommended: 'We recommend that you use Identity Center to provide console access to a person. With Identity Center, you can centrally manage user access to their AWS accounts and cloud applications.'
 - ☒ I want to create an IAM user: 'We recommend that you create IAM users only if you need to enable programmatic access through access keys, service-specific credentials for AWS CodeCommit or Amazon Keyspaces, or a backup credential for emergency account access.'
- Console password:** Two radio button options are present:
 - ☒ Autogenerated password: 'You can view the password after you create the user.'
 - ☐ Custom password: 'Enter a custom password for the user.' Below this is a password input field with masked characters and a 'Show password' checkbox.
- Users must create a new password at next sign-in - Recommended:** A checkbox is checked. A note below it says: 'Users automatically get the IAMUserChangePassword policy to allow them to change their own password.'
- Footer note:** A blue-bordered box contains the text: 'If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. Learn more'.

At the bottom right of the console, there are 'Cancel' and 'Next' buttons.

Set Permissions

- Permissions options → **"Attach policies directly"**
- Under **"Permissions policies"**, search and select: **AmazonEC2FullAccess** & **AmazonS3FullAccess**
- Click **"Next"** → **"Create user"**

Step 1
Specify user details

Step 2
Set permissions

Step 3
Review and create

Step 4
Retrieve password

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☐ Add user to group
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ Copy permissions
Copy all group memberships, attached managed policies, and inline policies from an existing user.

☒ Attach policies directly
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (2/1399) [Create policy](#)

Choose one or more policies to attach to your new user.

Search: [X](#) Filter by Type: 1 match

☒	Policy name ?	Type	Attached entities
☒	AmazonEC2FullAccess	AWS managed	0

► Set permissions boundary - optional

[Cancel](#) [Previous](#) [Next](#)

Step 1
Specify user details

Step 2
Set permissions

Step 3
Review and create

Step 4
Retrieve password

Review and create

Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name Omer	Console password type Autogenerated	Require password reset Yes
-------------------	--	-------------------------------

Permissions summary

Name ?	Type	Used as
AmazonEC2FullAccess	AWS managed	Permissions policy
AmazonS3FullAccess	AWS managed	Permissions policy
IAMUserChangePassword	AWS managed	Permissions policy

Generate Credentials

- Click on the newly created user
- Go to the **"Security credentials"** tab
- Under **"Access keys"**, click **"Create access key"**
- Choose **"Command Line Interface (CLI)"**
- Download the CSV file containing Access Key ID and Secret Access Key

The above steps are required to use AWS CLI. With the ID and Access Key, a user will be able to configure AWS CLI.

Logging in to the IAM user

- If the password is generated automatically, and the user is asked to reset it on first login, noting down the generated password is crucial. We can reset the password later on, but that will be one extra step.

Step 1
Specify user details

Step 2
Set permissions

Step 3
Review and create

Step 4
Retrieve password

Retrieve password

You can view and download the user's password below or email users instructions for signing in to the AWS Management Console. This is the only time you can view and download this password.

Console sign-in details [Email sign-in instructions](#)

Console sign-in URL
<https://734846753465.signin.aws.amazon.com/console>

User name
Omer

Console password
qHW7V2*W [Hide](#)

[Cancel](#) [Download .csv file](#) [Return to users list](#)

- With the account ID, username, and one-time password, log in to the IAM user.

Amazon Web Services Sign-In

https://eu-north-1.signin.aws.amazon.com/oauth?client_id=arn%3Aaws%3Asignin%3A%3A%3Ac...

IAM user sign in

Account ID or alias [\(Don't have?\)](#)
734846753465

☐ Remember this account

IAM username
Omer

Password
qHW7V2*W

☐ Show Password [Having trouble?](#)

[Sign in](#)

Amazon Lightsail

Lightsail is the easiest way to get started on AWS

[Learn more »](#)

- Change one-time/old password using a new one.

Provide feedback English ▾

aws

Password reset ⓘ

Your account (734846753465) password has expired or requires a reset.

To continue, please verify your old and set a new password for Omer (not you?).

Old Password

☒ Show Password

New Password

Confirm New Password

☒ Show Password Matches

[Confirm Password Change](#)

[Sign in to a different account](#)

© 2025 Amazon Web Services, Inc. or its affiliates. All rights reserved.

- Once logged in, the console of an IAM user will look like the image attached below.

aws Search [Alt+S] Europe (Stockholm) Account ID: 7348-4675-3465 Omer

Console Home Info

[Reset to default layout](#) [+ Add widgets](#)

Recently visited Info

No recently visited services

Explore one of these commonly visited AWS services.

[EC2](#) [S3](#) [Aurora and RDS](#) [Lambda](#)

[View all services](#)

Applications (0) Info

Region: Europe (Stockholm)

Select Region: eu-north-1 (Current Region) [Find applications](#)

Name	Description	Region	Orig
Access denied to servicecatalog:ListApplications			

[Diagnose with Amazon Q](#)

[Go to myApplications](#)

Welcome to AWS

[Getting started with AWS](#)

Learn the fundamentals and find valuable

AWS Health Info

Cost and usage Info

Current month: Access denied

Cost breakdown: Access denied

Forecasted month end

CloudShell Feedback

© 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

Verification:

- Try using EC2 Instance and an S3 bucket.

EC2

aws [Search] [Alt+S] Europe (Stockholm) Account ID: 7348-4675-3465 Omer

EC2 > Instances > Launch an instance

Success
Successfully initiated launch of instance (i-0d455b935539892e6)

Launch log

Initializing requests	✓ Succeeded
Launch initiation	✓ Succeeded

S3

aws [Search] [Alt+S] Europe (Stockholm) Account ID: 7348-4675-3465 Omer

Amazon S3 > Buckets

Successfully created bucket "omer-iam-user-s3-buc-buc"
To upload files and folders, or to configure additional bucket settings, choose [View details](#).

General purpose buckets All AWS Regions Directory buckets

General purpose buckets (2) Info

[Refresh](#) [Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Buckets are containers for data stored in S3.

Name	AWS Region	Creation date
omer-iam-user-s3-buc-buc	Europe (Stockholm) eu-north-1	October 13, 2025, 03:24:54 (UTC+05:30)
s3-endpoint-bucket-nv-test	US East (N. Virginia) us-east-1	October 13, 2025, 00:04:58 (UTC+05:30)

Account snapshot Info
Updated daily
[View dashboard](#)
Storage Lens provides visibility into storage usage and activity trends.

External access summary - new Info
Updated daily
External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

CloudFront:

aws [Search] [Alt+S] Global Account ID: 7348-4675-3465 Omer

CloudFront > Distributions

CloudFront

Distributions

Policies
Functions
Static IPs
VPC origins
What's new

SaaS

Multi-tenant distributions
Distribution tenants

Telemetry

Monitoring
Alarms

Failed to load distributions: User: arn:aws:iam::734846753465:user/Omer is not authorized to perform: cloudfront:ListDistributions because no identity-based policy allows the cloudfront:ListDistributions action
[Diagnose with Amazon Q](#)

Distributions (0) Info

[Refresh](#) [Enable](#) [Disable](#) [Delete](#) [Create distribution](#)

Filter type: All distributions

ID	Status	Descrip...	Type	Domai...	Alterna...	Origins
No distributions You don't have any distributions. Create distribution						

2. Create one Group in IAM and Assign Read access for EC2.

Purpose: Manage permissions for multiple users efficiently; allow monitoring/viewing without modification.

STEPS TO CREATE A GROUP IN IAM AND ASSIGN ONLY READ ACCESS TO EC2

Create Group

- Navigate to **IAM** → **"User groups"** → **"Create group"**
- Group name: **EC2-ReadOnly-Group**
- Click **"Next"**

Attach Policy

- Search for **AmazonEC2ReadOnlyAccess**
- Check the policy
- Click **"Create group"**

Create user group

Name the group

User group name

Enter a meaningful name to identify this group.

EC2-ReadOnly-Group

Maximum 128 characters. Use alphanumeric and '+=, @, _' characters.

Add users to the group - *Optional* (1) [Info](#)

An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.

Search

☐	User name ?	Groups	Last activity	Creation time
☐	Omer	0	18 minutes ago	20 minutes ago

Attach permissions policies - *Optional* (1/1079) [Info](#)

You can attach up to 10 policies to this user group. All the users in this group will have permissions that are defined in the selected policies.

Filter by Type				
Search	ec2re	All types	1 match	
☒	Policy name	Type	Used as	Description
☒	 AmazonEC2ReadOnlyAccess	AWS managed	None	Provides read only access to Amazon EC2 via the AWS Mana...

Cancel

Create user group

✓ EC2-ReadOnly-Group user group created.

[View group](#)

User groups (1) [Info](#)

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users.

Search

☐	Group name	Users	Permissions	Creation time
☐	EC2-ReadOnly-Group	0	Defined	Now

3. Create a new user with the name DevOps and add it to the group created in task 2.

Purpose: Provide a team member – DevOps, with EC2 read-only permissions through group membership; simplify permission management.

STEPS TO CREATE A NEW USER WITH NAME – DEVOPS, AND ADD THE USER TO ALREADY EXISTING GROUP – EC2-READONLY-GROUP

Create User

- Navigate to **IAM** → **"Users"** → **"Create user"**
- Username: **DevOps**
- Click **"Next"**

Add to Group

- Select **"Add user to user group"**
- Select **EC2-ReadOnly-Group**
- Click **"Create user"**

User details

User name

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)

☒ **Provide user access to the AWS Management Console - optional**
If you're providing console access to a person, it's a [best practice](#) to manage their access in IAM Identity Center.

Are you providing console access to a person?

User type

☐ **Specify a user in Identity Center - Recommended**
We recommend that you use Identity Center to provide console access to a person. With Identity Center, you can centrally manage user access to their AWS accounts and cloud applications.

☒ **I want to create an IAM user**
We recommend that you create IAM users only if you need to enable programmatic access through access keys, service-specific credentials for AWS CodeCommit or Amazon Keyspaces, or a backup credential for emergency account access.

Console password

☐ **Autogenerated password**
You can view the password after you create the user.

☒ **Custom password**
Enter a custom password for the user.

- Must be at least 8 characters long
- Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and symbols ! @ # \$ % ^ & * () _ + - (hyphen) = [] { } ' "

☒ **Show password**

☐ **Users must create a new password at next sign-in - Recommended**
Users automatically get the [IAMUserChangePassword](#) policy to allow them to change their own password.

Info If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user.
[Learn more](#)

[Cancel](#) [Next](#)

IAM > Users > Create user

Step 1
Specify user details

Step 2
Set permissions

Step 3
Review and create

Step 4
Retrieve password

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☒ **Add user to group**
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ **Copy permissions**
Copy all group memberships, attached managed policies, and inline policies from an existing user.

☐ **Attach policies directly**
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

User groups (1/1)

☒

Group name

Users

Attached policies

Created

☒

EC2-ReadOnly-Group

0

AmazonEC2ReadOnlyAc...

2025-10-13 (9 minut...

▶ Set permissions boundary - optional

Cancel

Previous

Next

IAM > Users > DevOps

Identity and Access Management (IAM)

Dashboard

▼ Access management

User groups

Users

Roles

Policies

Identity providers

Account settings

Root access management New

▼ Access reports

Access Analyzer

Resource analysis New

Unused access

Analyzer settings

Credential report

DevOps Info

Delete

Summary

ARN
arn:aws:iam::734846753465:user/DevOps

Console access
Enabled without MFA

Access key 1
[Create access key](#)

Created
October 13, 2025, 03:43 (UTC+05:30)

Last console sign-in
Never

Permissions

Groups (1)

Tags

Security credentials

Last Accessed

Permissions policies (1)

Remove

Add permissions

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type
All types

☐

Policy name

Type

Attached via

☐

AmazonEC2ReadOnlyAccess

AWS managed

Group EC2-ReadOnly-Group

Verification:

aws

[Alt+S]

Europe (Stockholm)

Account ID: 7348-4675-3465

DevOps

EC2 > Instances

EC2

Dashboard

AWS Global View

Events

▼ Instances

Instances

Instance Types

Instances (1) Info

Connect

Instance state

Actions

Launch instances

All states

Instance state = running

Clear filters

☐

Name

Instance ID

Instance state

Instance type

Status check

Alarm status

☐

omer-ec2

i-0d455b935539892e6

Running

t3.micro

3/3 checks passed

View alarms

The user only has read-only permissions. That's the reason the user can see the already existing instance, but is unable to create a new user.

The screenshot shows the AWS Management Console interface. The top navigation bar includes the AWS logo, a search bar, and the account ID: 7348-4675-3465. The left sidebar shows the navigation menu with categories like EC2, Images, and Elastic Block Store. The main content area displays an error message for the EC2 instance **i-Od455b935539892e6**. The error message states: "Failed to stop the instance i-Od455b935539892e6. You are not authorized to perform this operation, User: arn:aws:iam::734846753465:user/DevOps is not authorized to perform: ec2:StopInstances on resource: arn:aws:ec2:eu-north-1:734846753465:instance/i-Od455b935539892e6 because no identity-based policy allows the ec2:StopInstances action. Encoded authorization failure message: 2ByoL2DZQtdfy91v0TVqsJyA2VY6_BQ2zccDolwdemddo0f78arRG1vHz7Zlrhb7b2G9c2MUzfOKpSMwbLB0Gt0Hy4PO_8PudOT0FMWIRlpU5nxZALUGUx523EzScdQjBx03vuM-SrpH5Y5m3MBsuOL3S2IsW_HFQ7zK1muX0Cz3T1blu_YGndTqMDDN5LfWnYRR-unS-F-f1xQeM8IsFIQKB82Nst6BdVdTzeEQ7Xe8s9vNIA815wJB7h_4A53E46rdOPTiCpoP-3VEf-pnfRZS_mCauvHorihzVnJu-Pwme2L_8Kk98aV-pJlGMnxrineL_RUT5HZUA1H_DkPynZsWVjzjwPWaCN_AK_xuCO6NUcOfdJySMlryCZbjWbhonQ8jvIDTKx40suzetEsUik1sAPqj9V_S7tAHQk9VJA6OYYPas2dCycArFXKyfXnr9NM8NxyWdX_Fk5dUJ66sZYCu8Jwx_8o5QyMoNF5-_jwPoL5rvSEQcMMbWP9ZUHLW2eyXxbRCCRw7Lq51zh-ZZNB_4YGtjmUNMxTF44PDmRx1Jz_hLd-V7iRbcsQ7HGfJAZFbtD5hj93JYSIEIA9laf4c8yq3tYt-gOFA_rCycBXGbvVkoYys4Fs4FTQcpl9PJVZDMFOlskrY9mR17xb0VHTK-T2a4IKiRR9H56u66AV08SclBhcrBvRVINhIFASNzEVGNimi4DnVip8d4b1qEhndDIRBea0NZ7Z7rdR_Mo_1". Below the error message, the instance details for **i-Od455b935539892e6 (omer-ec2)** are shown, including the Instance ID, Public IPv4 address (16.171.113.145), and Private IPv4 addresses (172.31.25.93).

The user doesn't have access to any other resource other than EC2

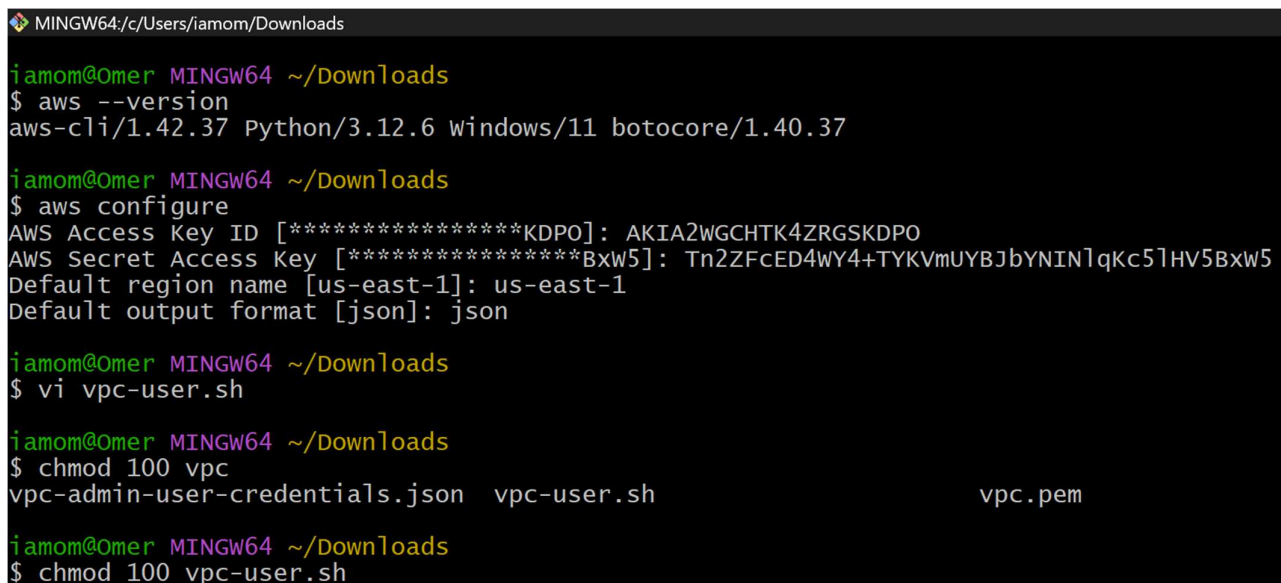
The screenshot shows the AWS Management Console interface for the Amazon S3 service. The top navigation bar includes the AWS logo, a search bar, and the account ID: 7348-4675-3465. The left sidebar shows the navigation menu with categories like Amazon S3, Amazon ElastiCache, and Amazon EMR. The main content area displays an error message for attempting to create a bucket. The error message states: "Error: Access Denied". Below the error message, the bucket details for **General purpose buckets (0)** are shown, including the Name, AWS Region, and Creation date.

4. Write a bash script to create an IAM user with VPC full access.

Purpose: Enable Infrastructure as Code (IaC) approach; reduce manual errors; speed up user provisioning.

STEPS TO CREATE A USER WITH A BASH SCRIPT AND GIVE FULL ACCESS TO VPC:

- Open Git Bash and check if AWS CLI is installed
\$ aws --version
- If installed, proceed with the next step. If not, install AWS CLI.
- Configure AWS CLI
\$ aws configure
- Enter the configuration details – Access Key ID, Secret Access Key, Region name, and Format type.
- Create a bash file – vpc-user.sh
\$ touch vpc-user.sh
\$ chmod 100 vpc-user.sh -> Important for script execution



```
MINGW64:/c/Users/iamom/Downloads
iamom@Omer MINGW64 ~/Downloads
$ aws --version
aws-cli/1.42.37 Python/3.12.6 Windows/11 botocore/1.40.37

iamom@Omer MINGW64 ~/Downloads
$ aws configure
AWS Access Key ID [*****KDPO]: AKIA2WGCHTK4ZRGSKDPO
AWS Secret Access Key [*****Bxw5]: 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
Default region name [us-east-1]: us-east-1
Default output format [json]: json

iamom@Omer MINGW64 ~/Downloads
$ vi vpc-user.sh

iamom@Omer MINGW64 ~/Downloads
$ chmod 100 vpc
vpc-admin-user-credentials.json  vpc-user.sh  vpc.pem

iamom@Omer MINGW64 ~/Downloads
$ chmod 100 vpc-user.sh
```

Add the following script to the vpc-user.sh file

```
#!/bin/bash

# Script: create-iam-vpc-user.sh
# Purpose: Automate IAM user creation with VPC full access
# Usage: ./create-iam-vpc-user.sh <username>

set -e # Exit on error

# Color codes for output
RED='\033[0;31m'
GREEN='\033[0;32m'
YELLOW='\033[1;33m'
```


NC='\033[0m' # No Color

Function to display usage

```
usage() {  
    echo "Usage: $0 <username>"  
    echo "Example: $0 vpc-admin-user"  
    exit 1  
}
```

Function to check if AWS CLI is installed

```
check_aws_cli() {  
    if ! command -v aws &> /dev/null; then  
        echo -e "${RED}Error: AWS CLI is not installed${NC}"  
        exit 1  
    fi  
    echo -e "${GREEN}✓ AWS CLI found${NC}"  
}
```

Function to create IAM user

```
create_user() {  
    local username=$1  
    echo -e "${YELLOW}Creating IAM user: $username${NC}"  
  
    if aws iam get-user --user-name "$username" 2>/dev/null; then  
        echo -e "${YELLOW}User $username already exists${NC}"  
    else  
        aws iam create-user --user-name "$username"  
        echo -e "${GREEN}✓ User $username created successfully${NC}"  
    fi  
}
```

Function to attach VPC full access policy

```
attach_vpc_policy() {  
    local username=$1  
    echo -e "${YELLOW}Attaching VPC full access policy...${NC}"  
  
    aws iam attach-user-policy --user-name "$username" \  
        --policy-arn arn:aws:iam::aws:policy/AmazonVPCFullAccess  
  
    echo -e "${GREEN}✓ VPC full access policy attached${NC}"  
}
```

Function to create access keys

```
create_access_keys() {  
    local username=$1
```



```

local output_file="${username}-credentials.json"

echo -e "${YELLOW}Creating access keys...${NC}"

aws iam create-access-key --user-name "$username" \
    > "$output_file"

echo -e "${GREEN}✓ Access keys created${NC}"
echo -e "${GREEN}Credentials saved to: $output_file${NC}"
}

# Function to display summary
display_summary() {
    local username=$1
    echo -e "\n${GREEN}===== USER CREATION SUMMARY =====${NC}"
    echo -e "Username: ${GREEN}$username${NC}"
    echo -e "Access Level: ${GREEN}VPC Full Access${NC}"
    echo -e "Policies Attached: ${GREEN}AmazonVPCFullAccess${NC}"
    echo -e "Credentials File: ${GREEN}${username}-credentials.json${NC}"
    echo -e "${YELLOW} ⚠ Keep credentials file secure and share with user only${NC}"
    echo -e "${GREEN}===== ${NC}\n"
}

# Main script execution
main() {
    # Validate input
    if [[ $# -eq 0 ]]; then
        usage
    fi

    local username=$1

    # Validate username format
    if ! [[ $username =~ ^[a-zA-Z0-9._-]+$ ]]; then
        echo -e "${RED}Error: Username contains invalid characters${NC}"
        echo "Username must contain only alphanumeric characters, dots, hyphens, and underscores"
        exit 1
    fi

    echo -e "${GREEN}Starting IAM user creation process...${NC}\n"

    # Execute steps
    check_aws_cli
    create_user "$username"
    attach_vpc_policy "$username"

```

```

create_access_keys "$username"
display_summary "$username"
}

```

```

# Run main function
main "$@"

```

Run the script and give Username Input

```
$ ./vpc-user.sh <username>
```

```
$ ./vpc-user.sh techie
```

```

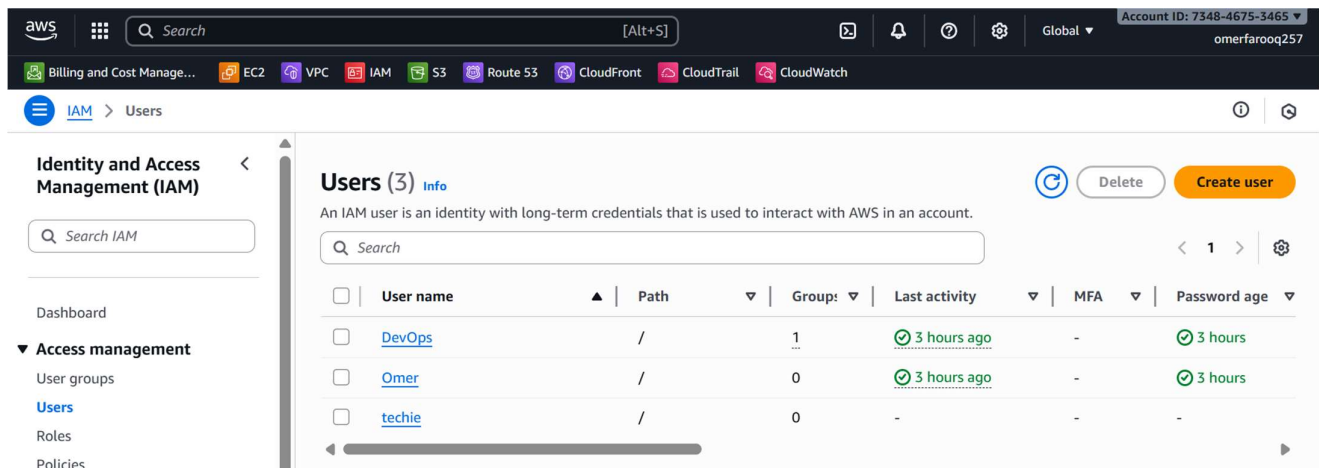
iamom@Omer MINGW64 ~/Downloads
$ ./vpc-user.sh techie
Starting IAM user creation process...

✓ AWS CLI found
Creating IAM user: techie
{
  "User": {
    "Path": "/",
    "UserName": "techie",
    "UserId": "AIDA2WGCHTK4ZRJ54LMPK",
    "Arn": "arn:aws:iam::734846753465:user/techie",
    "CreateDate": "2025-10-13T01:28:07Z"
  }
}
✓ User techie created successfully
Attaching VPC full access policy...
✓ VPC full access policy attached
Creating access keys...
✓ Access keys created
Credentials saved to: techie-credentials.json

===== USER CREATION SUMMARY =====
Username: techie
Access Level: VPC Full Access
Policies Attached: AmazonVPCFullAccess
Credentials File: techie-credentials.json
⚠ Keep credentials file secure and share with user only
=====

```

- Check the created user in IAM → Users

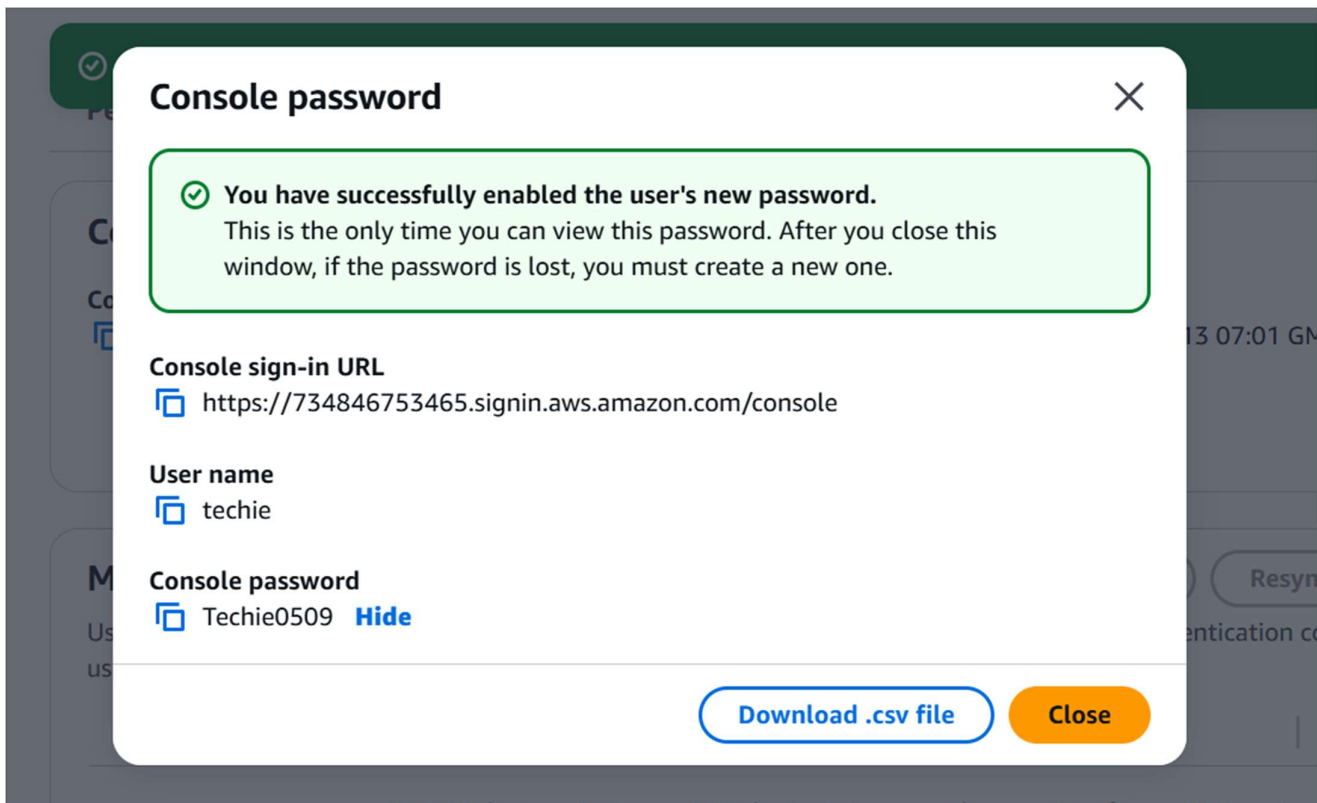


The screenshot shows the AWS IAM console interface. The left sidebar contains navigation links for Identity and Access Management (IAM), Access management, User groups, Roles, and Policies. The main content area displays the 'Users (3)' page, which includes a search bar and a table of users. The table lists three users: DevOps, Omer, and techie. The 'techie' user is highlighted, showing its path as '/', group as '0', and last activity as '3 hours ago'. The 'techie' user is also listed in the table with a path of '/', group of '0', and last activity of '3 hours ago'.

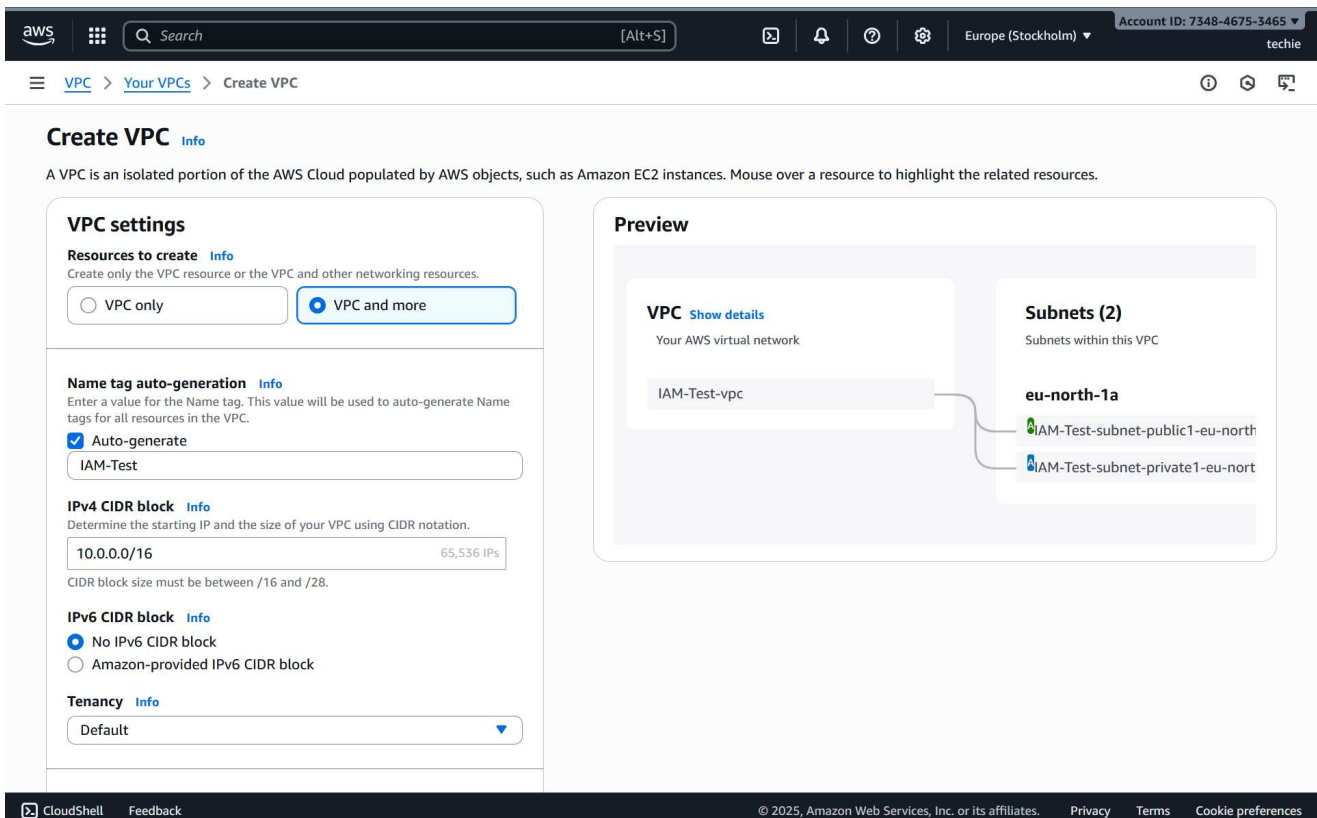
User name	Path	Group	Last activity	MFA	Password age
DevOps	/	1	3 hours ago	-	3 hours
Omer	/	0	3 hours ago	-	3 hours
techie	/	0	-	-	-

Login and Verification:

- Log in to the console using the username and password of the newly created **techie** user.



Navigate to VPC and check access



aws

Search

[Alt+S]

Europe (Stockholm)

Account ID: 7348-4675-3465

techie

VPC

Your VPCs

Create VPC

Create VPC resources

Create VPC workflow

Success

Details

✓ Create VPC: [vpc-0ad03a2a5c61be7cd](#)

✓ Enable DNS hostnames

✓ Enable DNS resolution

✓ Verifying VPC creation: [vpc-0ad03a2a5c61be7cd](#)

✓ Create subnet: [subnet-05645cda2b700341b](#)

✓ Create subnet: [subnet-081c2914b5f7f2a69](#)

✓ Create internet gateway: [igw-0043e5774cd1cefe0](#)

✓ Attach internet gateway to the VPC

✓ Create route table: [rtb-0823e811be6bc4a4e](#)

✓ Create route

✓ Associate route table

✓ Create route table: [rtb-051d88a441d3686f5](#)

✓ Associate route table

✓ Verifying route table creation

View VPC

CloudShell

Feedback

© 2025, Amazon Web Services, Inc. or its affiliates.

Privacy

Terms

Cookie preferences

vpc-0ad03a2a5c61be7cd / IAM-Test-vpc

Actions

Details

Info

VPC ID

[vpc-0ad03a2a5c61be7cd](#)

DNS resolution

Enabled

Main network ACL

[acl-026ace9aad0bcc84a](#)

IPv6 CIDR (Network border group)

—

State

Available

Tenancy

default

Default VPC

No

Network Address Usage metrics

Disabled

Block Public Access

Off

DHCP option set

[dopt-0e8c7ae3668fd8ddb](#)

IPv4 CIDR

10.0.0.0/16

Route 53 Resolver DNS Firewall rule groups

Failed to load rule groups

DNS hostnames

Enabled

Main route table

[rtb-0425f8a399fbc1f68](#)

IPv6 pool

—

Owner ID

[734846753465](#)

Resource map

CIDRs

Flow logs

Tags

Integrations

Resource map

Info

VPC

Your AWS virtual network

IAM-Test-vpc

Subnets (2)

Subnets within this VPC

eu-north-1a

✓ IAM-Test-subnet-public1-eu-north-1a

✗ IAM-Test-subnet-private1-eu-north...

Route tables (3)

Route network traffic to resources

IAM-Test-rtb-public

rtb-0425f8a399fbc1f68

IAM-Test-rtb-private1-eu-north-1a

Network Connections (1)

Connections to other networks

IAM-Test-igw

Check the VPC from the root user.

aws

Search

[Alt+S]

Europe (Stockholm)

Account ID: 7348-4675-3465

omerfaroog257

Billing and Cost Manage...

EC2

VPC

IAM

S3

Route 53

CloudFront

CloudTrail

CloudWatch

VPC

Your VPCs

VPC dashboard

<

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Your VPCs (2)

Info

Last updated less than a minute ago

Actions

Create VPC

Find VPCs by attribute or tag

☐

Name

☐

IAM-Test-vpc

[vpc-0ad03a2a5c61be7cd](#)

Available

Off

10.0.0.0/16

☐

—

[vpc-0179019d50398dd7a](#)

Available

Off

172.31.0.0/16

5. Create an IAM policy to access EC2 for a specific user in a specific region only.

Purpose: Enforce security boundaries; prevent accidental/malicious access to resources in other regions; and cost control.

STEPS TO CREATE A CUSTOMER MANAGED POLICY TO RESTRICT EC2 ACCESS TO A SPECIFIC REGION AND APPLY THE POLICY TO A USER.

Navigate to IAM Policies

- Go to **IAM** → **"Policies"** → **"Create policy"**

Select Policy Editor

- Choose **"JSON"** tab

Enter Policy JSON

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "EC2AccessUSEast1Only",
      "Effect": "Allow",
      "Action": [
        "ec2:*"
      ],
      "Resource": "*",
      "Condition": {
        "StringEquals": {
          "aws:RequestedRegion": "us-east-1"
        }
      }
    },
    {
      "Sid": "DenyAllOutsideUSEast1",
      "Effect": "Deny",
      "Action": [
        "ec2:*"
      ],
      "Resource": "*",
      "Condition": {
        "StringNotEquals": {
          "aws:RequestedRegion": "us-east-1"
        }
      }
    }
  ],
  "Resource": "*"
}
```

```

    "Sid": "AllowEC2DescribeGlobally",
    "Effect": "Allow",
    "Action": [
        "ec2:DescribeRegions",
        "ec2:DescribeAvailabilityZones"
    ],
    "Resource": "*"
}
]
}

```

IAM > Policies > Create policy

Step 1
Specify permissions

Step 2
Review and create

Specify permissions [Info](#)

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Policy editor

```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "EC2AccessUSEastOnly",
6       "Effect": "Allow",
7       "Action": [
8         "ec2:*"
9       ],
10      "Resource": "*",
11      "Condition": {
12        "StringEquals": {
13          "aws:RequestedRegion": "us-east-1"
14        }
15      }
16    },
17    {
18      "Sid": "DenyAllOutsideUSEast",
19      "Effect": "Deny",
20      "Action": [
21        "ec2:*"
22      ],
23      "Resource": "*",
24      "Condition": {
25        "StringNotEquals": {
26          "aws:RequestedRegion": "us-east-1"
27        }
28      }
29    }
30  ]
31 }

```

+ Add new statement

JSON Ln 40, Col 1

Security: 0 Errors: 0 Warnings: 0 Suggestions: 0

Visual JSON Actions

Edit statement

Select a statement

Select an existing statement in the policy or add a new statement.

+ Add new statement

5677 of 6144 characters remaining

Cancel Next

IAM > Policies > Create policy

Step 1
Specify permissions

Step 2
Review and create

Review and create [Info](#)

Review the permissions, specify details, and tags.

Policy details

Policy name
Enter a meaningful name to identify this policy.

EC2-US-East-1-Only

Maximum 128 characters. Use alphanumeric and '+=,._@-' characters.

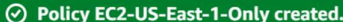
Description - optional
Add a short explanation for this policy.

This policy allows access to EC2 only in the North Virginia region when attached.

Maximum 1,000 characters. Use alphanumeric and '+=,._@-' characters.

Review and Create

- Name: EC2-US-East-1-Only
- Click "Create policy"

 [View policy](#) 

Policies (1400) [Info](#)

A policy is an object in AWS that defines permissions.

Filter by Type Customer managed ▼ 4 matches < 1 > 

	Policy name ▲	Type ▼	Used as ▼	Description
☐	 EC2-US-East-1-Only	Customer managed	None	This policy allows access to EC2 only in.
☐	 s3crr_for_lounge-tem...	Customer managed	Permissions policy (1)	-
☐	 s3crr_for_lounge-tem...	Customer managed	Permissions policy (1)	-
☐	 s3replicate_for_loung...	Customer managed	Permissions policy (1)	-

Attach to User

- Go to Users → Select user
- "Add permissions" → "Attach policies directly"
- Search for EC2-US-East-1-Only
- Click "Attach policies"

 IAM > Users Info  

Dashboard

▼ Access management

- User groups
- Users**
- Roles
- Policies
- Identity providers
- Account settings
- Root access management [New](#)

Users (1/2) [Info](#)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

< 1 > 

	User name ▲	Path ▼	Group: ▼	Last activity ▼	MFA ▼	Password age ▼
☒	DevOps	/	1	-	-	✓ 42 minutes
☐	Omer	/	0	✓ 1 hour ago	-	✓ 1 hour

 IAM > Users > DevOps Info  

▼ Access management

- User groups
- Users**
- Roles
- Policies
- Identity providers
- Account settings
- Root access management [New](#)

Permissions policies (1)  [Remove](#) [Add permissions ▼](#)

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type All types ▼ < 1 > 

	Policy name  ▲	Type ▼	Attached via 
☐	  AmazonEC2ReadOnlyAccess	AWS managed	Group EC2-ReadOnly-Group

▼ Access reports

Step 1

Add permissions

Step 2

Review

Add permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

**Add user to group**

Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

**Copy permissions**

Copy all group memberships, attached managed policies, inline policies, and any existing permissions boundaries from an existing user.

**Attach policies directly**

Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1/1400)



Filter by Type

Customer ma... ▼

4 matches

< 1 >



Policy name



Type



Attached e...

[EC2-US-East-1-Only](#)

Customer managed

0

DevOps

Info

Delete

Summary

ARN

arn:aws:iam::734846753465:user/DevOps

Created

October 13, 2025, 03:43 (UTC+05:30)

Console access

Enabled without MFA

Last console sign-in

Never

Access key 1

[Create access key](#)**Permissions**

Groups (1)

Tags

Security credentials

Last Accessed

Permissions policies (2)

Permissions are defined by policies attached to the user directly or through groups.



Remove

Add permissions ▼

Filter by Type

All types

< 1 >



Policy name



Type



Attached via

[AmazonEC2ReadOnlyAccess](#)

AWS managed

Group [EC2-ReadOnly-Group](#)[EC2-US-East-1-Only](#)

Customer managed

Directly

Verification

aws

[Alt+S]

United States (N. Virginia) ▼

Account ID: 7348-4675-3465 ▼

DevOps

EC2 > Instances > Launch an instance

i Q

It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices

Take a walkthrough

Do not show me this message again.

Launch an instance

Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Info

Name

e.g. My Web Server

Add additional tags

Application and OS Images (Amazon Machine Image)

Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

[Quick Start](#)**Summary****Number of instances** Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.9.2...[read more](#)
ami-052064a798f08fd3

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

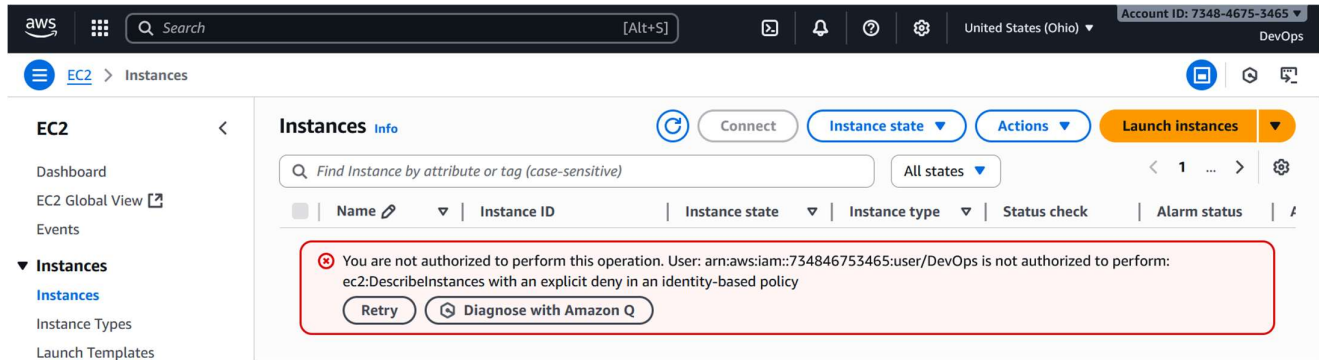
Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

- Try to launch any instance in us-east-1 and any other region.
- If everything is done right, you will not be allowed access to EC2 in another region.



6. We have two accounts, Account A and Account B. The Account A user should access the S3 bucket in Account B.

Purpose: Facilitate secure multi-account architecture; enable teams to work across organizational accounts without sharing credentials.

STEPS TO PERFORM THE TASK: